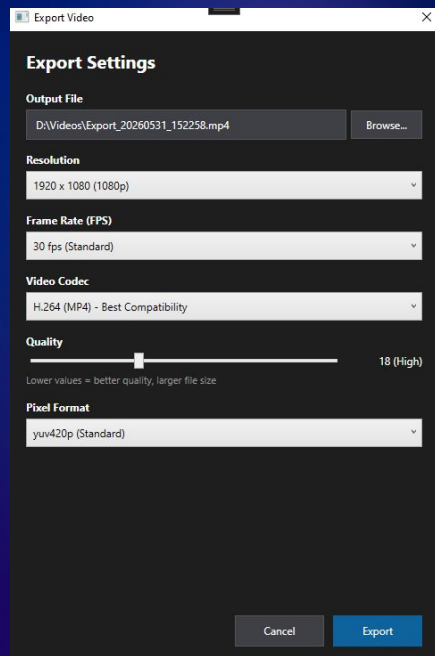
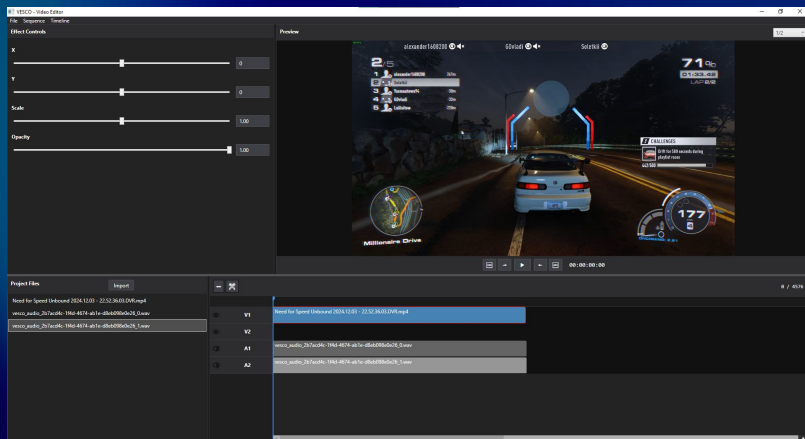


Софтуер за Видеообработка



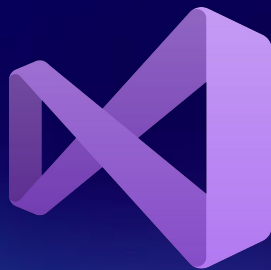
Дипломант: Стоян Иванов
Дипломен ръководител: Марио Бенов

ФУНКЦИОНАЛНИ ИЗИСКВАНИЯ

1. Вмъкване (импортиране) на файлове
2. Работа с видео и аудио клипове
3. Редактиране на видео клипове
4. Визуализация в реално време
5. Управление на основни параметри на видео клиповете
6. Експортиране

СРЕДСТВА ЗА РАЗРАБОТКА

core



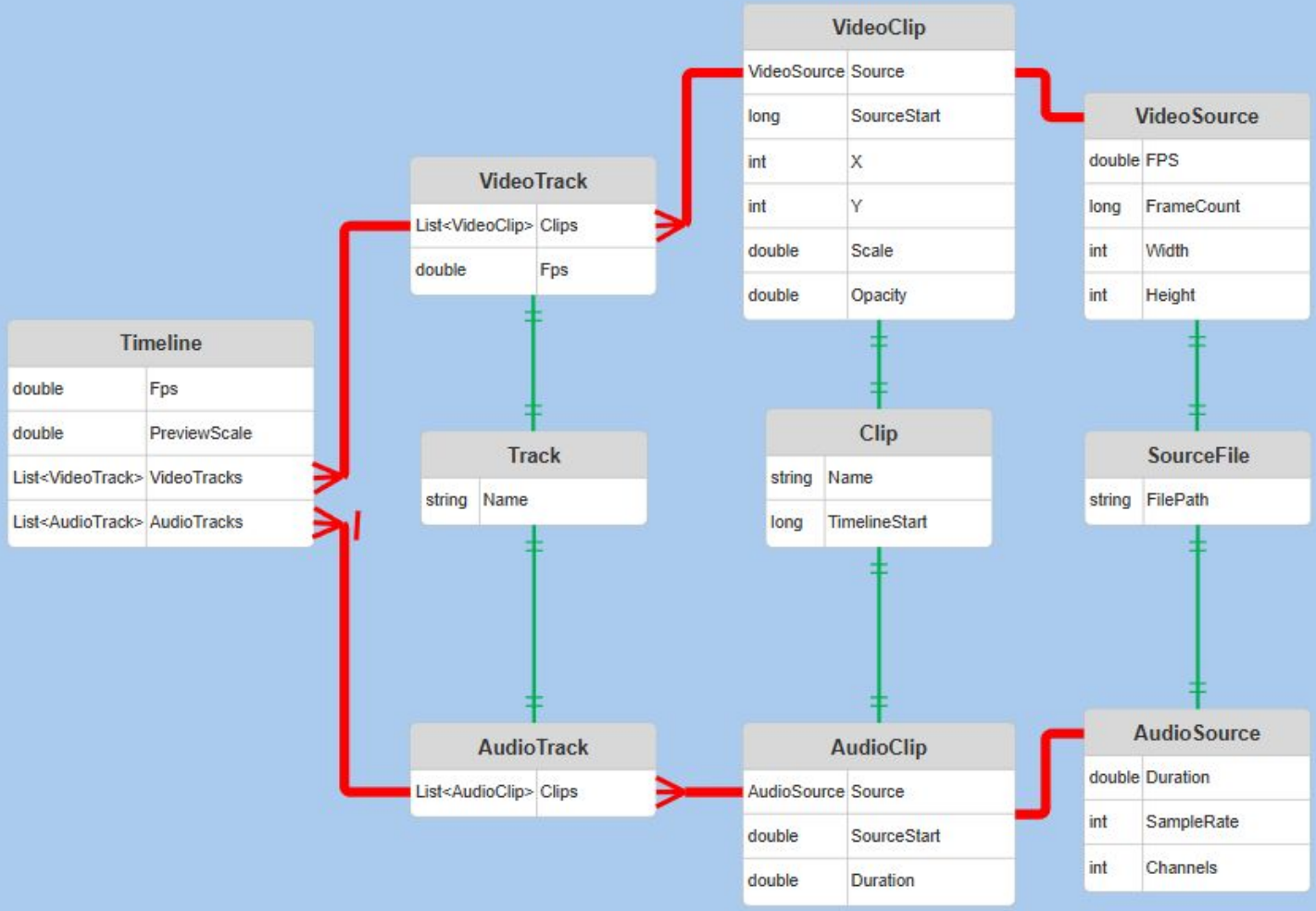
video & audio



АРХИТЕКТУРА



M



УПРАВЛЯЮЩИЕ МОДУЛИ

ClipDrawManager

ClipSelectionManager

PlayheadController

SnapManager

TimelineController

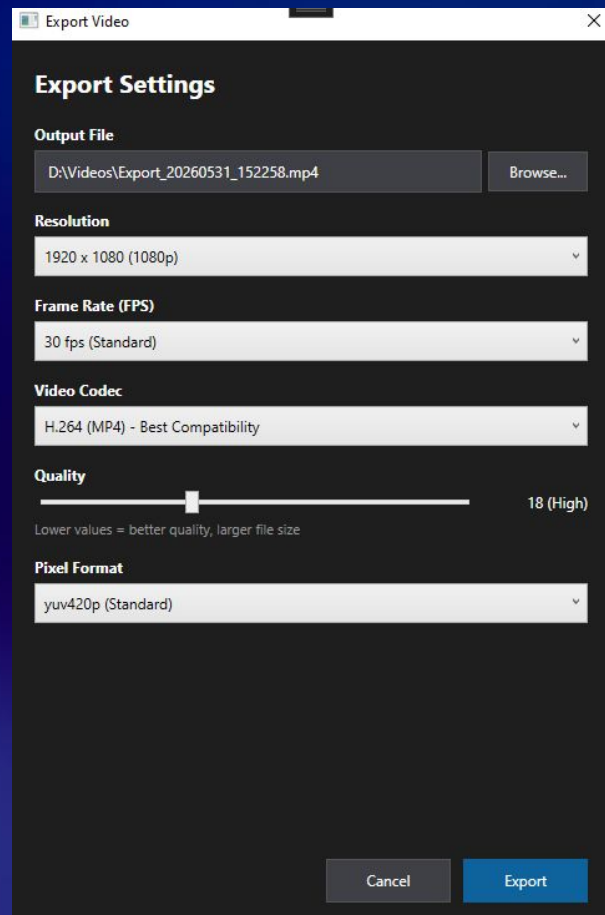
ToolManager

Track Manager

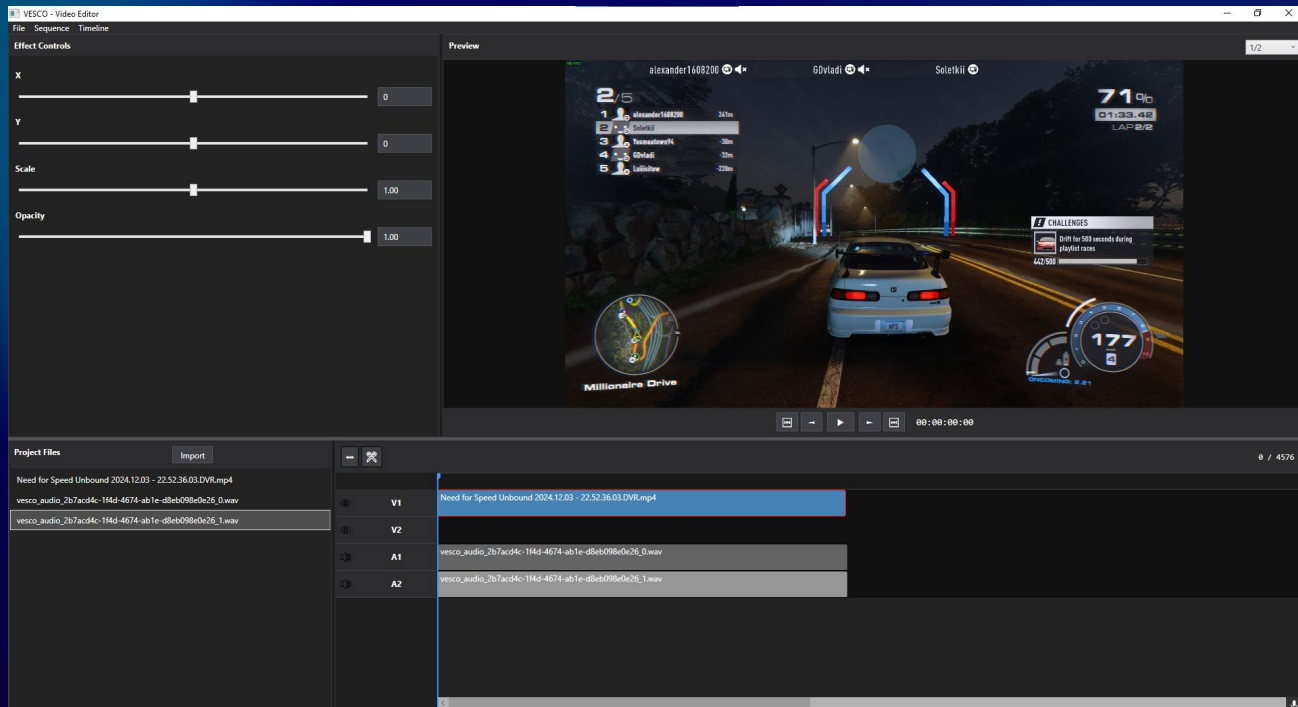
ЕКСПОРТИРАНЕ

VideoRenderer

1. RenderVideo
2. Създава Temp папка в %appdata%
3. Запазва кадри от Timeline като .png
4. ExportAudio
5. EncodeWithFFmpeg



ПОТРЕБИТЕЛСКИ ИНТЕРФЕЙС



DEMO

БЪДЕЩО РАЗВИТИЕ

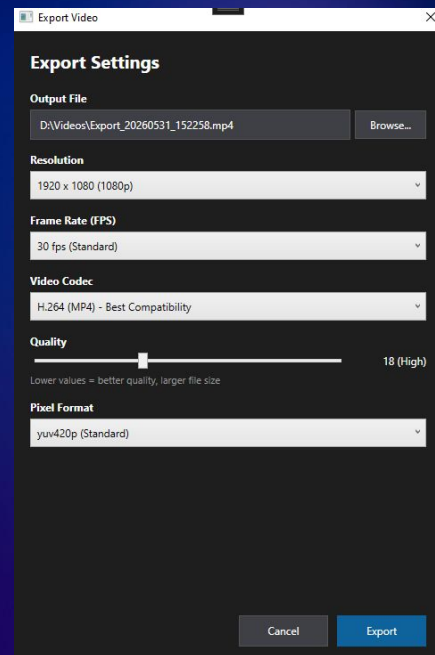
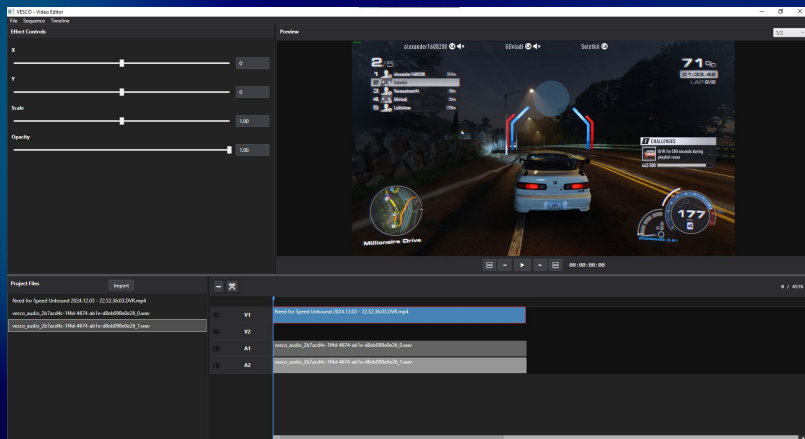


Audio Editing



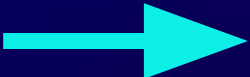
Video Decoding
Class

Благодаря за вниманието!



Очаквам вашите въпроси!

1. Как оптимизацията на PreviewScale може да се съчетае с FrameCache?

Index $\rightarrow$ Bitmap  Index & PreviewScale $\rightarrow$ Bitmap

0.50000000000002 (double)  500 (int)

2. Какво представлява Mat и как работи Cv2.CvtColor?

Mat е основният OpenCV контейнер за изображение.

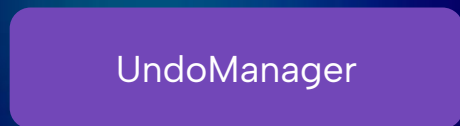
“Mat is basically a class with two data parts: the matrix header (containing information such as the size of the matrix, the method used for storing, at which address is the matrix stored, and so on) and a pointer to the matrix containing the pixel values (taking any dimensionality depending on the method chosen for storing) . The matrix header size is constant, however the size of the matrix itself may vary from image to image and usually is larger by orders of magnitude.”

- OpenCV 2.4 13.7 documentation

`Cv2.CvtColor(src, dst, code)`

`enum ColorConversionCodes` - `GRAY2BGR`, `BGRA2BGR`, ...

3. Как биха могли да се реализират undo и redo функционалности?



- undo()
- redo()

AddClipCommand, RemoveClipCommand,
MoveClipCommand, AddTrackCommand...

